

| Assumption tested                   | If low value occurs                      | If baseline value occurs                                      | If high value occurs                                  | Implication for Scenario A: Full replacement                                               | Implication for Scenario B: Component repair and replacement                                                               | Related risk classification                       |
|-------------------------------------|------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Frame life                          | Frame lasts only 30 years                | Frame lasts around 40 years                                   | Frame lasts around 50 years                           | Replacement may be more justified if frame life is short                                   | Repair strategy becomes stronger if the frame remains usable longer                                                        | Lifecycle trade-off or technological lock-in      |
| Glazing life                        | Glazing needs replacement after 20 years | Glazing needs replacement around 25 years                     | Glazing lasts up to 35 years                          | Full replacement may be chosen if glazing performance drops early                          | Component glazing replacement may avoid replacing the full frame                                                           | Component replacement or implementation gap       |
| Hardware replacement interval       | Hardware wears after 10 years            | Hardware replacement around 15 years                          | Hardware lasts around 20 years                        | Full replacement is not justified if only hardware fails                                   | Scenario B is stronger if hardware can be replaced independently                                                           | Premature replacement                             |
| Repair events                       | Only one repair event needed             | Two repair events needed                                      | Three repair events needed                            | Full replacement may appear simpler if many repairs are needed                             | More repair events may still be beneficial if they avoid full-window replacement, but transport and labour must be checked | Lifecycle trade-off                               |
| Transport distance                  | 50 km                                    | 100 km                                                        | 300 km                                                | Full replacement burden increases if transport and removal distances are high              | Repair burden increases if multiple service trips are long-distance                                                        | Burden shifting                                   |
| Spare-part availability             | Parts remain available long-term         | Parts available for normal service period but uncertain later | Parts become difficult to find after older system age | Full replacement may become more likely when parts are unavailable                         | Scenario B fails if compatible parts cannot be sourced                                                                     | Implementation gap                                |
| Component access                    | Good access                              | Partly accessible                                             | Poor access                                           | Full replacement may be chosen if repair access is blocked                                 | Scenario B depends strongly on physical access to repairable components                                                    | Implementation gap                                |
| End-of-life route                   | Favourable recycling                     | Mixed or uncertain recovery                                   | Poor recovery or downcycling                          | Scenario A depends heavily on how the removed full window is treated                       | Scenario B delays full-window end of life but still creates smaller waste flows                                            | Burden shifting                                   |
| Operational performance improvement | Small improvement from replacement       | Moderate improvement                                          | Large improvement                                     | Full replacement becomes more plausible when operational improvement is large and verified | Repair remains plausible if performance requirements are still met or glazing can be upgraded                              | Lifecycle verification                            |
| Client preference for new product   | Low preference for replacement           | Moderate preference                                           | Strong preference for newer or more modern windows    | Full replacement may occur before technical failure                                        | Scenario B may be undermined by user-driven replacement decisions                                                          | Premature replacement or user-driven obsolescence |